FundFlow — Global Mutual Fund ETL Pipeline

Complete Technical Documentation and Implementation Report
Evidence snapshot: 28 October 2026 · local project/API evidence

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1. Executive Summary

FundFlow is a multi-country mutual-fund catalogue and NAV ETL application. The inspected project combines Python connectors and ETL, PostgreSQL persistence, a Node.js/Express API, and a React/Vite dashboard. Its local API snapshot reports **183,169 fund records**, **283,590 NAV observations**, **24 configured markets**, **51 registered sources**, and **30 sources marked Integrated**.

The separate SEC holdings snapshot previously verified in the same local database reports **17,487 US funds with holdings and 5,433,511 filed position rows**. Coverage is not global: no Brazil holdings rows were present in that snapshot, despite Brazil having 37,875 fund records and 252,065 NAV observations. Optional market metrics belong to holding_metrics, distinct from filed portfolio positions. Production hosting was not verified; the owner reported deleting the prior Render services. These figures describe a dated local snapshot, not a live deployed service.

Evidence status. “Verified” below means confirmed from repository code, the local API snapshot, or the prior read-only database query as identified in the Evidence Notes. Source registry metadata can be candidate or unverified: being listed or marked integrated does not prove each URL is reachable or that every field is licensed for reuse. The database could not be freshly queried during this report generation because its local PostgreSQL connection was unavailable; the holdings counts therefore retain their prior snapshot provenance.

2. Project Overview

FundFlow consolidates fund master records and net asset values from public and official datasets into a common database, exposes that information through a REST API, and presents it in a dashboard. The user-facing functions observed in source include overview statistics, country/source views, fund search/filtering, fund details, NAV history, portfolio holdings, and ETL run history/control.

The architecture is appropriate for a data-oriented portfolio project: source-specific adapters isolate upstream formats, database uniqueness constraints support repeatable ingestion, and the API keeps frontend requests away from direct SQL access. No user portfolio or trading workflow is documented as implemented.

3. Country Coverage

The country API returns **25 registry codes**; the dashboard statistic reports **24 configured countries** because Spain (ES) is excluded from assignment-country count by the API query. The country table below follows the country endpoint, while “configured” follows /api/v1/stats. Counts are records stored under a country code, not estimates of each market's total mutual-fund universe.

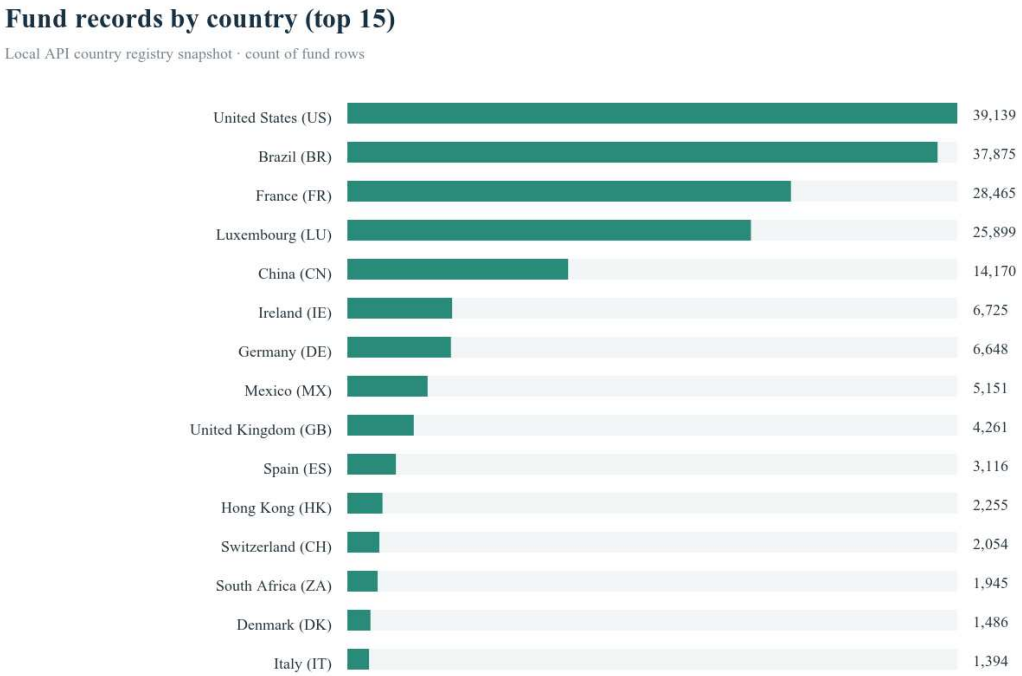


Figure 1. Countries sorted by stored fund records. Values are generated from the country API snapshot; the chart shows the 15 largest markets.

Code	Country	Region	Sources	Ingested sources	Funds	NAV observations	Registry status
AE	United Arab Emirates	Other markets	1	0	0	0	not_integrated
AU	Australia	Asia-Pacific	3	2	97	5	integrated
BR	Brazil	Other markets	1	1	37,875	252,065	integrated

Code	Country	Region	Sources	Ingested sources	Funds	NAV observations	Registry status
CA	Canada	North America	2	1	373	373	integrated
CH	Switzerland	Europe	3	2	2,054	94	integrated
CN	China	Asia-Pacific	1	1	14,170	0	integrated
DE	Germany	Europe	3	2	6,648	248	integrated
DK	Denmark	Europe	2	1	1,486	0	integrated
ES	Spain	Europe	1	1	3,116	0	integrated
FR	France	Europe	2	2	28,465	90	integrated
GB	United Kingdom	Europe	4	3	4,261	286	integrated
HK	Hong Kong	Asia-Pacific	2	1	2,255	0	integrated
IE	Ireland	Europe	2	1	6,725	0	integrated
IT	Italy	Europe	3	2	1,394	68	integrated
JP	Japan	Asia-Pacific	1	0	0	0	not_integrated
KR	South Korea	Asia-Pacific	1	0	0	0	not_integrated
LU	Luxembourg	Europe	4	3	25,899	1,944	integrated
MX	Mexico	North America	2	1	5,151	3,486	integrated

NL	Netherlands	Europe	3	2	1,390	84	integrated
RU	Russia	Other markets	1	0	0	0	not_integrated
SE	Sweden	Europe	2	1	726	0	integrated
SG	Singapore	Asia-Pacific	1	0	0	0	not_integrated
TW	Taiwan	Asia-Pacific	1	0	0	0	not_integrated
US	United States	North America	3	2	39,139	24,847	integrated
ZA	South Africa	Other markets	2	1	1,945	0	integrated

The accompanying FundFlow_Country_Coverage_New.csv also includes holdings counts. Holdings are currently evidenced only for the US in the prior snapshot; zero elsewhere means no holding rows observed in that snapshot, not proof the source has no holdings data. The United States totals are 39,139 funds, 24,847 NAV observations in the country API (the prior DB-specific US query counted 24,847 rows and 24,826 funds with NAV), and the SEC holdings totals stated above.

4. Data Sources and Verification

The source catalogue has **51 source records** across public regulators, government agencies, exchanges, and data portals. Examples include the US SEC EDGAR Form N-PORT dataset and Brazil's CVM fund information / CDA holdings data. The inventory CSV retains each catalogue record's organization, URLs, access method, format, licensing fields, verification status, and integration status.

Registry measure	Count	Interpretation
Registered sources	51	Catalogue rows, not necessarily active integrations
Marked Integrated	30	Registry state only; connector success is separately logged
Not Started	21	No implemented integration status recorded
Usable fund data confirmed	30	Registry verification classification
Unavailable	20	Current registry verification classification
Accessible, no fund data	1	Accessible source with no usable fund records confirmed

Licensing metadata is not fully resolved: 48 entries show Unknown, 2 Permitted, and 1 Permission Required in the snapshot. Commercial reuse must be checked against the source terms before external redistribution. Source-specific references are linked in FundFlow_Data_Source_Inventory_New.csv; the report references authoritative entry points in Section 18.

5. ETL Architecture and Execution

Python CLI commands select source connectors, retrieve source payloads, parse records, map them to canonical schemas, validate field types, and persist accepted rows. The loader uses PostgreSQL conflict handling and the schema declares unique keys for source-native funds and fund/date/currency NAV observations. This supports repeatable upserts; it does not make every upstream feed complete or guarantee a full historical backfill.

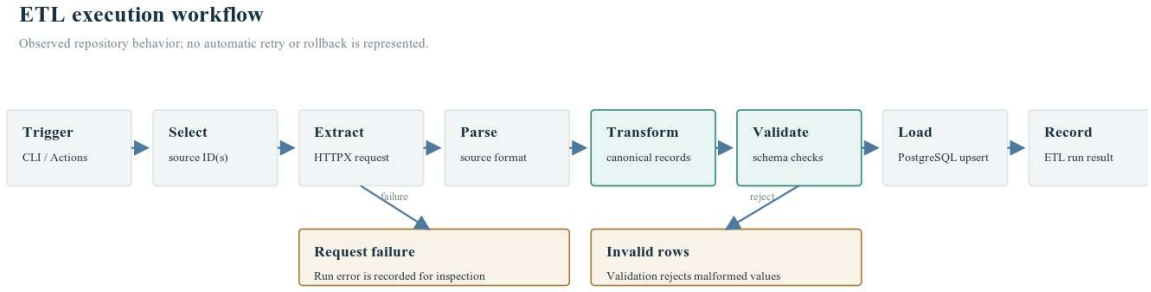


Figure 2. Current workflow based on repository code. Validation and request failures are recorded/rejected as applicable; automatic retry/rollback is not depicted because it was not verified.

GitHub Actions defines a manual dispatch and weekly schedule (0 2 * * 0 UTC). The workflow applies migrations, syncs the source catalogue, runs available connectors, syncs SEC holdings, and performs count checks. Workflow configuration does not prove that a run completed successfully. The most recent API/ETL run in the saved snapshot is python-277, succeeded, finished 2026-10-27T21:30:22.561Z.

6. Data Model and Database

The PostgreSQL schema separates catalogue entities, point-in-time observations, filed portfolio positions, optional position metrics, and run logs. The current Alembic head recorded in the prior inspection is 0006_portfolio_holding_details.

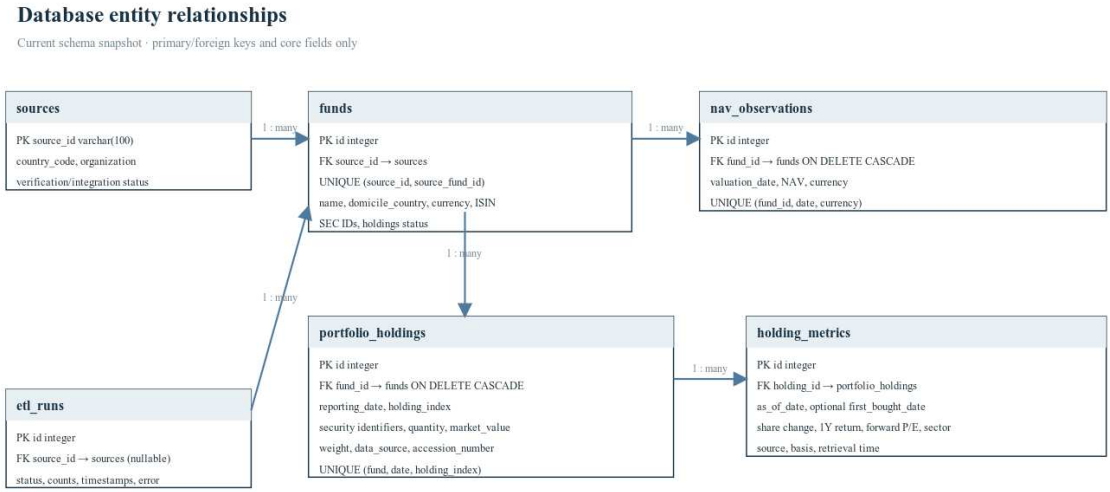


Figure 3. Core tables, selected columns, key constraints, and relationship cardinalities. Full columns are documented in FundFlow_Database_Inventory_New.md.

Table	Purpose	Key integrity rules
sources	Registry of source feeds	source_id primary key
funds	Fund metadata	(source_id, source_fund_id) unique; source FK
nav_observations	Daily/periodic NAV points	(fund_id, valuation_date, currency) unique
portfolio_holdings	Filed portfolio positions by report date	(fund_id, reporting_date, holding_index) unique; fund FK with cascade
holding_metrics	Optional sourced/calculated metrics	(holding_id, data_source, as_of_date) unique; holding FK with cascade

Table	Purpose	Key integrity rules
etl_runs	Python connector run outcomes	optional source FK, row counts, timestamps and error
api_etl_runs	API-triggered run log	API UUID, source selection, status and result details
api_source_checks	Source connectivity checks	source-linked check history
alembic_version	Migration revision state	Alembic-managed revision marker

7. Dashboard and Fund Profiles

The React dashboard calls API endpoints rather than embedding fund totals. It presents summary counts, country coverage, recent runs, recent funds, source status, and navigation to fund explorer, API explorer, and ETL control. Fund detail code supports fund metadata, NAV history, reporting-date selection, holdings rows, and sourced optional metrics. The frontend includes a responsive layout and theme styling, but a dedicated screenshot file could not be saved with the available UI capture tool; see the screenshots folder README for the capture steps. No screenshot is fabricated or embedded.

8. NAV Coverage

The local API snapshot reports **283,590 NAV observations** across **57,135 funds** from the prior database query. The recorded date span was **2011-12-06 through 2026-10-27**. Country NAV counts are listed in the table and CSV. Country data is uneven: Brazil contributes 252,065 rows, Luxembourg 1,944, Mexico 3,486, and the US 24,847. Several countries have fund records but no NAV rows, so a nonzero fund count must not be read as NAV coverage.

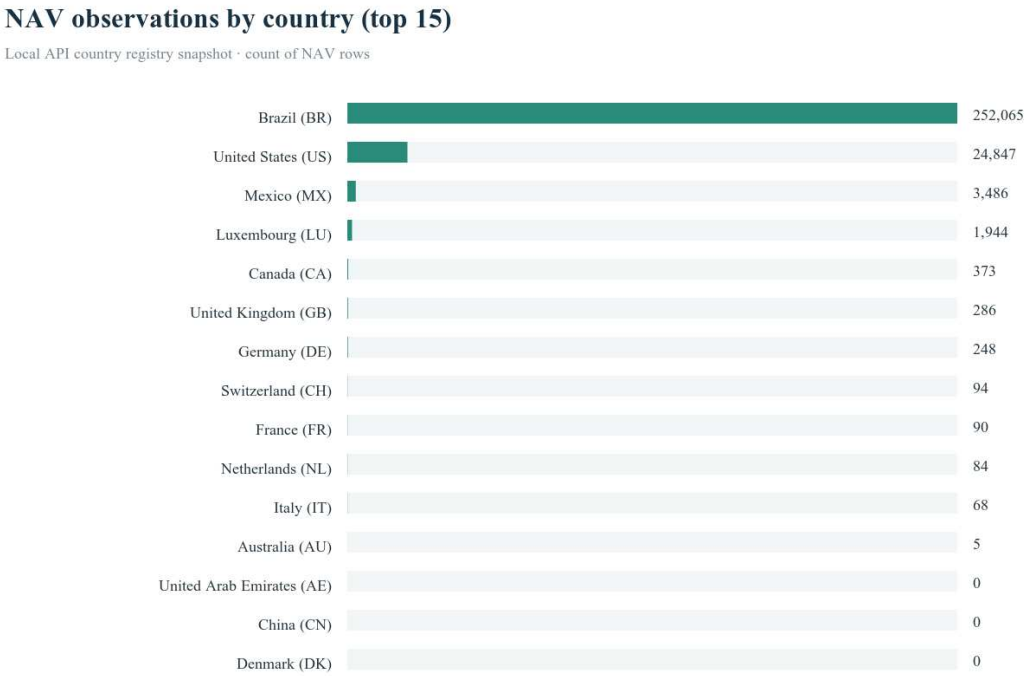


Figure 4. NAV observations by country, sorted descending, top 15; generated from the same endpoint snapshot as the coverage table.

9. Portfolio Holdings

Filed holdings are stored per fund and reporting date, preserving historical portfolios. Fields include security name, ticker, ISIN/CUSIP, quantity, market value/currency, portfolio weight, filing accession, source URL, source identifier, and retrieval timestamp. Optional metrics such as first bought date, share change, one-year return, forward P/E, and sector are stored separately in holding_metrics, with a source and as-of date. The prior local snapshot contained 355 metric rows; share-change values were present, while first-bought date, one-year return, forward P/E, and sector were not populated in that check. Missing optional values remain null; the UI should not infer or invent them.

The holdings endpoint supports report-date filtering and bounded pagination. Holdings availability should be interpreted separately from fund and NAV availability. Data originates from official filings where available; market metrics require their own verified provider and provenance.

10. US Mutual Fund Holdings

The US collection uses SEC EDGAR Form N-PORT data (sec_edgar_nport). The prior local database query verified **17,487 US funds**, **5,433,511 position rows**, across **19 reporting dates** from **2025-06-30 to 2026-08-31**. US fund records total 39,139, leaving **21,652** without holdings rows in that snapshot. That gap may reflect filing/reporting eligibility, mapping coverage, fund type, or update timing; it is not evidence that those funds have no portfolio.

A coalesced key count of 264,440 was previously computed using CUSIP, ISIN, ticker, then security name. This is only a proxy for distinct securities and is not a canonical issuer/security count. Prior row-level quality check found no null security name, quantity, market value, or portfolio weight in the holdings rows; optional metrics can still be null. SEC reports are periodic filings, not real-time portfolios.

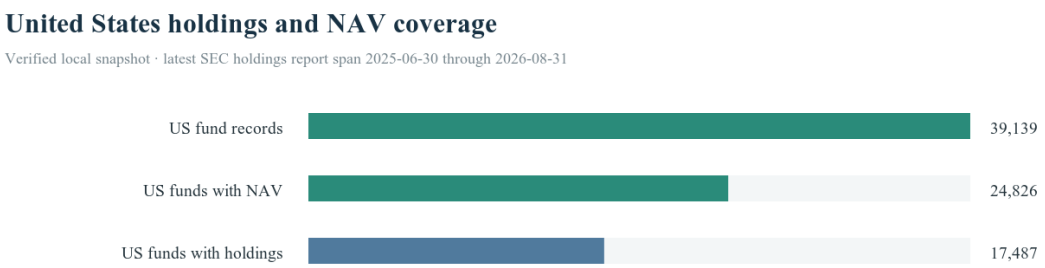


Figure 5. US funds with NAV and holdings compared with US fund records. Distinct security key is intentionally omitted because the available identifier fallback is not a canonical instrument identity.

11. REST API

The Express API performs parameterized PostgreSQL queries, validates request parameters, bounds pagination, and routes errors through middleware. Database health is a separate route from general process health. The endpoint inventory and representative response contracts are in FundFlow_API_Inventory_New.md.

Key endpoints include GET /health, GET /api/v1/health/database, GET /api/v1/stats, GET /api/v1/countries, GET /api/v1/sources, GET /api/v1/funds, GET /api/v1/funds/:id, GET /api/v1/funds/:id/nav, GET /api/v1/funds/:id/holdings, and ETL run history/start routes. Fund listing accepts filters including country, currency, text, category/type/status, holdings status, ISIN, NAV availability, sort, page, and limit. The holdings limit is bounded (up to 5,000); NAV history is bounded separately.

12. Technology and Configuration

Layer	Technology verified in repository
ETL	Python, HTTPX, Pydantic, SQLAlchemy, Alembic
API	Node.js, Express, PostgreSQL pg, Zod
Frontend	React 18, Vite, Lucide icons
Database	PostgreSQL; local compose configuration is present
Automation	GitHub Actions workflow, scheduled and manual
Runtime configuration	DATABASE_URL, USER_AGENT, HTTP_TIMEOUT_SECONDS, API/admin settings

Secret values are not included. Local and hosted database targets must be configured independently; copying local credentials/data into a report is neither required nor safe.

13. Deployment and Automation

The local development topology includes Vite on port 5173, API on port 3001, and PostgreSQL configured by environment. GitHub Actions is configured to run ETL against the database URL stored as a repository secret. Earlier Render hosting was used, but the user later reported deleting those services; current production endpoints, database, frontend/backend hosting, and deployment health could not be verified for this report.

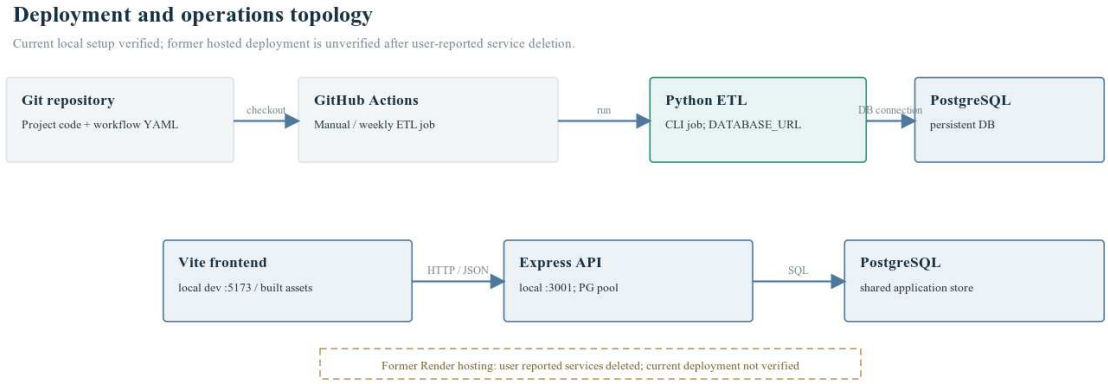


Figure 6. Current local/CI configuration; dashed hosted box explicitly marks the prior Render deployment as unverified.

14. Verified Statistics

Measure	Value	Evidence and caveat
Funds	183,169	Local API /stats, snapshot 2026-10-28
NAV observations	283,590	Local API /stats, same snapshot
Configured countries	24	Stats query excludes Spain; country API has 25 codes
Registered sources	51	Source catalogue API
Integrated sources	30	Registry state, not proof of live reachability
Funds with NAV	57,135	Prior direct local database query
US funds	39,139	Country API snapshot
US funds with NAV	24,826	Prior direct local database query
US holdings funds / rows	17,487 / 5,433,511	Prior direct local database query
Database size	2,630 MB (~2.57 GiB)	Prior read-only database query; not freshly rechecked
Python ETL run rows	232 total; 222 succeeded, 10 failed	Prior database query; success is historical
API ETL runs	5 succeeded	Prior database query

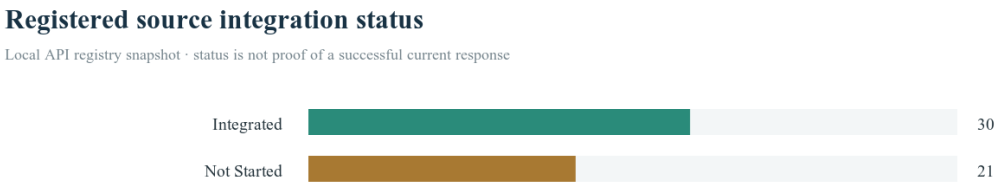


Figure 7. Registry counts generated from the 51-source API snapshot. Do not interpret as source uptime or successful current data extraction.

15. Charts and Diagrams

All charts in this report are generated as SVG from the saved API country/source JSON snapshots or previously queried database counts. SVG preserves vector quality in the PDF. The architecture and workflow diagrams are hand-laid SVG derived from the inspected code and workflow. The ER diagram uses actual schema table names and constraints. Source files and figure outputs are saved separately under reports/diagrams/FundFlow_New/ and reports/figures/FundFlow_New/.

16. Quality, Security, and Limitations

The schema and loader use unique constraints/upserts to avoid duplicate source-native fund/NAV/holding rows. Source metadata exposes verification and licensing limitations. Request validation, parameterized queries, pagination bounds, and error middleware are implemented. Secrets are excluded from artifacts.

Limitations: coverage varies by country and by data type; source registry presence is not proof of usable data; license/commercial rights are unresolved for most sources; NAV histories can be sparse; filings are periodic; holdings are US-heavy; Brazil holdings were unavailable in the observed database; optional financial metrics are sparse and provider-dependent. Production is unverified after reported hosting deletion. Fresh DB checks could not run during this report build, and screenshots could not be persisted using the available UI capture interface.

17. Test and Verification Results

The frontend production build passed using npm run build from frontend/. The API test suite was attempted using npm test from api/: 2 tests passed and 15 tests could not start because the sandbox denied localhost binding (listen EPERM: operation not permitted 127.0.0.1); this is an environment limitation, not a passing test result. Local API health and DB connection could not be rechecked during report generation because the local API was not running and direct PostgreSQL connection failed. Existing API snapshot files and previously queried database results were retained with their provenance and caveats.

18. References and Evidence

1. SEC EDGAR Form N-PORT data sets: SEC data page.
2. SEC EDGAR developer resources and fair-access guidance: SEC developer resources.
3. Brazil CVM open data portal / fund datasets: CVM Dados Abertos.
4. PostgreSQL documentation: PostgreSQL current documentation.
5. Alembic documentation: Alembic migration guide.
6. React documentation: React.
7. Vite documentation: Vite.
8. Express documentation: Express.
9. GitHub Actions documentation: GitHub Actions.

The complete 51-source registry URLs and statuses are in FundFlow_Data_Source_Inventory_New.csv; country rollups are in FundFlow_Country_Coverage_New.csv. FundFlow_Report_Evidence_New.md distinguishes endpoint snapshots, repository inspection, prior DB queries, and checks that remain unavailable. URLs in the inventory are source catalogue claims; live reachability and legal usage should be independently rechecked before operational or commercial use.